



Using FlexiConc via CLiC

A case study on literary fiction

FlexiConc is a Python library developed to support concordance analysis for corpus linguists. CLiC incorporates FlexiConc functionality in a dedicated analysis tab. You can find general information on how FlexiConc is used in CLiC [here](#).

In this document, we will use FlexiConc to analyse literary fiction.

This document serves as an overview and illustrates all algorithms that we currently have available. You can either follow the exercises or look at individual algorithms to see examples of how to use them.

Our case study deals with body part nouns in 19th century English novels. It aims to find repeated patterns of language use. We use the 19C through the CLiC web app (Mahlberg et al. 2020).

The website can be accessed here: <https://cllc-fiction.com/>

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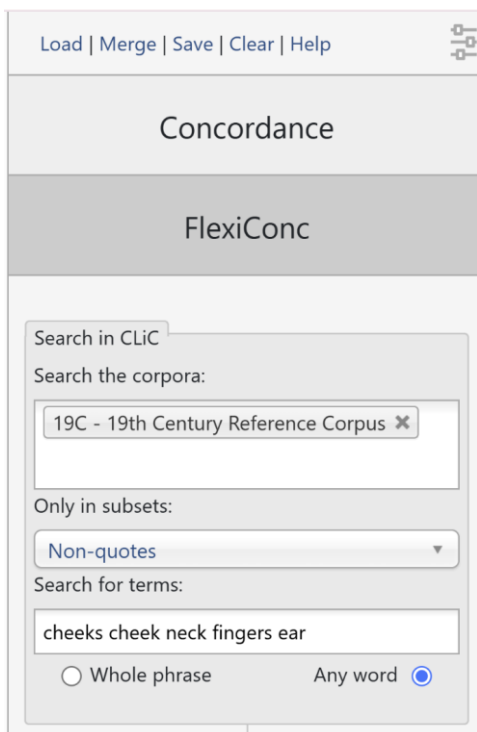
1. Running a concordance search in FlexiConc mode

Background: The concordance search is the basis for all subsequent steps like sorting or grouping. In the example, we search for a range of body part nouns in *non-quotes* – all parts of the 19C novels outside quotation marks. All further steps in this document are based on the same concordance created here. To use FlexiConc in CLiC, select the FlexiConc tab on the right side of the page.

Instructions:

1. Set the **corpus** to **19C**, **subset** **non-quotes**, and enter the **search terms** **cheeks cheek neck fingers ear**. The terms should be added with just a space as a separator (no commas or similar – Figure 1).
2. Select the **radio button** **Any word** to search the words separately rather than as a sequence (Figure 1).
3. Click the **confirmation button** to start searching the corpus (Figure 2).
4. You will see the resulting **concordance** in the main window (Figure 3).

Enter your search terms here:



The screenshot shows the FlexiConc interface. At the top, there are tabs for 'Concordance' and 'FlexiConc', with 'FlexiConc' being the active tab. Below the tabs, there is a search area. It includes a 'Search in CLiC' label, a 'Search the corpora:' section with a dropdown menu showing '19C - 19th Century Reference Corpus', and an 'Only in subsets:' section with a dropdown menu showing 'Non-quotes'. Below these, there is a 'Search for terms:' section with a text input field containing 'cheeks cheek neck fingers ear'. At the bottom of the search area, there are two radio buttons: 'Whole phrase' and 'Any word', with 'Any word' being selected.

Figure 2: Entering search terms.

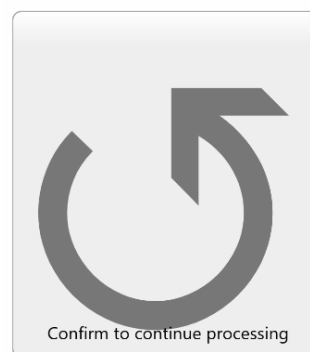


Figure 1: Confirmation button.

This is what your result should look like:

Showing 1 to 50 of 1,575 entries,					
64b4590					
ID	Left	Node	Right	Book	In bk.
0	tinge of crimson flushed the waxen whiteness of her	cheeks;	not one shadow of brown redeemed the pale insipi	LadyAud	
1	past nine o'clock, singing a little Scotch melody, her	cheeks	tinged with as delicate a pink as the pale hue	LadyAud	
2	's face; the pretty, roseate flush faded out from her	cheeks,	and left them waxen white, and angry flashes lighte	LadyAud	
3	unless, flaxen hue which generally accompanies pink	cheeks	and white eyelashes. She had been a rustic beauty,	LadyAud	
4	ing about him, kissed the dirty tears from his faded	cheeks.	"Don't be sorry for me, gran'pa," he said; "I am	LadyAud	
5	The unnatural color still burnt like a flame in her	cheeks;	the unnatural light still glittered in her eyes. The ex	LadyAud	
6	ter nor half the folks as are growed up." "I Maggie's	cheeks	began to flush with triumphant excitement. She tho	mill	
7	lavishing so many warm kisses on it that the waxen	cheeks	had a wasted, unhealthy appearance. "Did you eve	mill	
8	red with the roundest of eyes above the reddest of	cheeks;	but was now entirely defaced by a long career of	mill	
9	much alike as goslings,--a lad with light-brown hair,	cheeks	of cream and roses, full lips, indeterminate nose and	mill	
10	any, and they ate together and rubbed each other's	cheeks	and brows and noses together, while they ate, with	mill	

Figure 3: Concordance view.

2. A first overview: random sort

Background:

By default, concordance lines are displayed in the following order:

1. Concordance lines are shown in the order in which the search terms were entered.
2. If there is more than one result, the hits are sorted alphabetically based on the name of the book they appear in.
3. If there is more than one result per search term and book, the hits are sorted based on their position within the book.

Random sorting does precisely what the name suggests: the lines are displayed in a sequence that does not follow a systematically motivated order. That said, the order is not entirely random either: The **optional seed** number ensures that every time someone runs the same 'random sort' step on the same data, they will be presented with the same ordering. This helps to make sure that results are **reproducible**.

Instructions:

1. **Scroll down** in the FlexiConc panel until you see the **add algorithm** button below the tree symbol (Figure 4).
2. **Scroll down** in the selection menu to find the **random sort** function and click on it.
3. The algorithm will run immediately. Keep the **optional seed** that is set to **42** by default (Figure 5).

Find and use Random Sort like this:

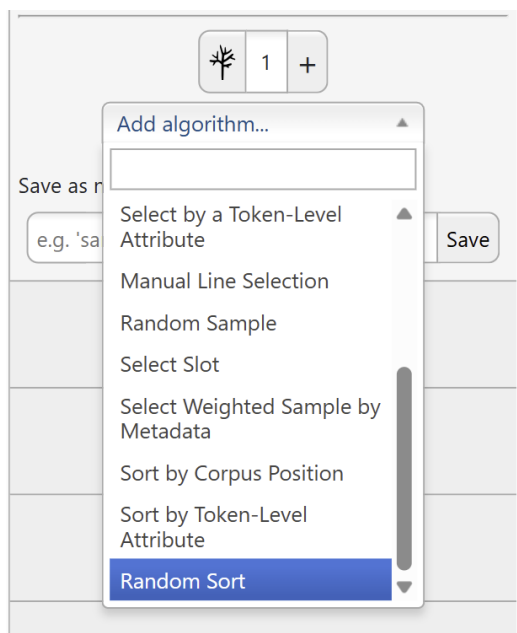


Figure 4: Finding the *Random Sort* algorithm.

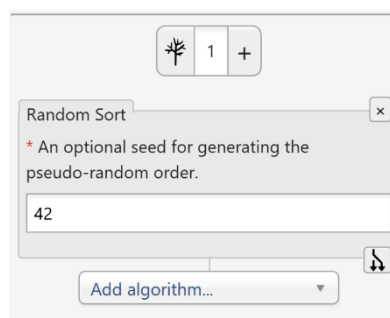


Figure 5: Seed for *Random Sort*

This is what your result should look like:

change over the counter, in taking which he caught her	fingers	and held them. There was a slight struggle and titter
uttering these three terrible _musts_, Klesmer lifted up three long	fingers	in succession. "In sum, you have not been called upon
the music was her only refuge from herself. Sometimes her	fingers	touched the notes with a lingering fondness--a soft, plaintive
appalled. His vows of austerity and celibacy echoed in his	ear;	his thirst after holiness--had it been quenched at so
the circle. Madame Carolina first presented her delicate and faintly-rouged	cheek	to the hump-backed Crown Prince, who scarcely raised his eyes
end over the lady's heart and the other to his	ear,	and listened carefully. ¶ When he had done he says to
her window, and then went to assist Mrs. Thornton's trembling	fingers.	¶ From some cause or other, there was a pause of
kind of nightmare; I felt the fiend's grasp in my	neck	and could not free myself from it; groans and cries
which Nature had gifted Lady Crawley were those of pink	cheeks	and a white skin, and as she had no sort
the high-cornered outline of an early Gothic shield; and a	neck	which was smooth and round as a cylinder. The lower

Figure 6: Concordance arranged by *Random Sort*

3. Adding information: annotations

Background: *Annotation* refers to the automatic analysis of the concordance lines. In CLiC's FlexiConc mode, several types of annotation are available:

- SpaCy POS tags
- Sentence transformers
- SpaCy embeddings
- TF-IDF embeddings

! Note: Annotations are different from algorithms: **you will not see immediate changes** to your concordance after running an annotation step. What annotation does is to enable the use of tags or embeddings *in certain algorithms*.

In the following two subsections, we will show you examples of the two most central annotations: POS tagging with spaCy and embeddings with sentence transformers (Reimers & Gurevych 2019).

3.1 POS tagging

Background:

PoS (part-of-speech) tags are a cornerstone of corpus linguistics. A POS tag, such as *ADJ* for adjectives, represents the morpho-syntactic category of a token. This information can be used in place of tokens in all algorithms that operate on *token-level attributes*.

In the FlexiConc integration of the CLiC web app, the concordance can be annotated with POS tags from the NLP library spaCy (<https://spacy.io/usage/linguistic-features>). POS tagging is supported as an annotation layer that is added to the concordance. After doing so, we can use the tags in all algorithms that rely on token-level attributes.

In the following example, we add POS tags using the **en_core_web_md** model. SpaCy also offers different models and additional grammatical information such as dependency. Also, note that the labels used for POS tagging may differ depending on the models used. For details, please refer to the spaCy documentation.

Instructions:

1. Find the **add annotation** menu.
2. Select **annotate with spaCy POS tags** (figure 7).
 ! FlexiConc supports various types of annotations that produce different results. Make sure to choose *SpaCy POS tags* and not, e.g., *SpaCy embeddings*.
3. Select the **model** *en_core_web_md* (figure 8)
 ! CLiC currently offers two English and one German tagging model. They are sorted alphabetically, with the German one (*de_core_news_sm*) active by default. When working on English data, make sure to change this selection to one of the *en_* models.
4. To add POS tagging, **select the attribute** *pos_* from the dropdown menu of spaCy token attributes (figure 9).
5. Click **confirm** in the concordance window.

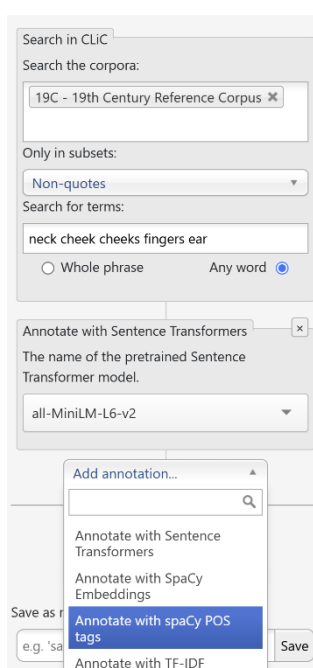
Find and annotate spaCy POS tags like this:

Figure 7: *Annotate with spaCy POS tags.*

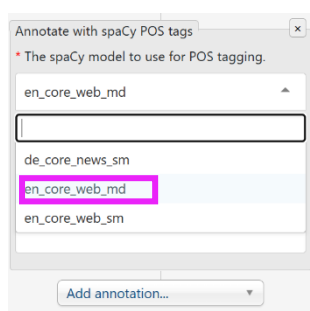


Figure 8: *spaCy models:* *de_* is for German, *en_* for English.

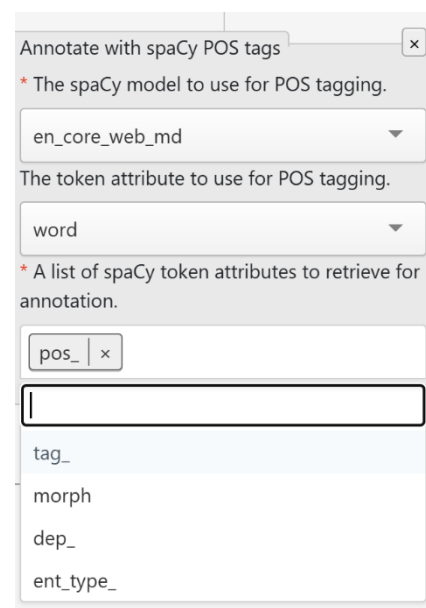


Figure 9: Token attributes for annotation

We are not showing you a screenshot of the concordance output here, because it will not change from adding an annotation. Instead, the result of this annotation is that you can now use the *pos_* attribute in algorithms that require token-level attributes. For example, *Select by Token-Level String Attribute* will now allow you to select lines containing adjectives, as shown in [Step 4.2](#).

3.2 Embeddings

Background:

Embedding annotations form the basis for *clustering* algorithms (cf. section 5). Clustering is used in FlexiConc to form partitions based on the similarity between concordance lines. To perform clustering, the desired annotation must be added beforehand.

As illustrated in the second screenshot below, embedding algorithms can use a **context window** with upper and lower bounds. For instance, you can set the upper bound to -1 to calculate similarities based on the left context.

Each type of embedding annotation represents a different method for determining similarity scores. While TF-IDF and spaCy embeddings are relatively established, sentence transformers (Reimers & Gurevych 2019) are a more recent method that draws on data from a pre-trained transformer model.

In this example, we add annotations based on sentence transformers.

! Note: As with the annotation for POS tags in the previous step, this annotation will not produce a visible difference to your concordance view. Instead, it is the prerequisite for the algorithm [Flat Clustering by Embeddings](#).

Instructions:

1. Click on the **add annotation** button below the query menu (Figure 10).
2. Select **annotate with Sentence Transformers** (Figure 10).
3. Click **confirm** in the concordance window.
4. Available options are shown in Figure 11- for our running example, leave them as-is.

Find and annotate sentence transformer embeddings like this:

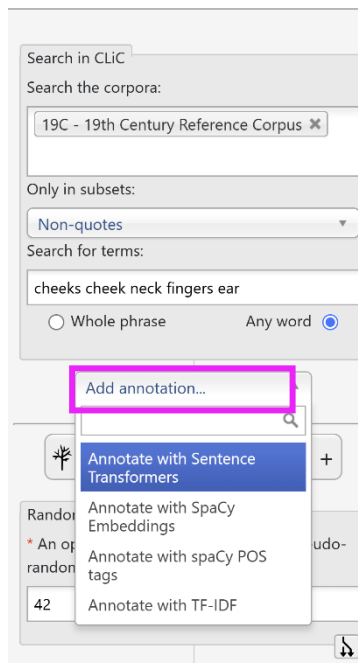


Figure 10: *Annotate with Sentence Transformers*

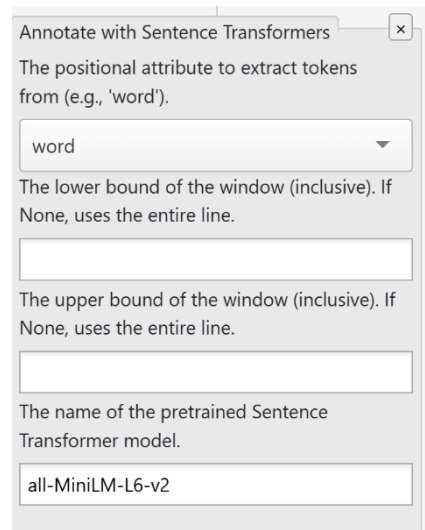


Figure 11: Parameter settings for *Annotate with Sentence Transformers*

As mentioned in the note above, the concordance output will not change from adding an annotation. Instead, the result of embedding annotations is that you can now use the resulting similarities for clustering with the algorithm *Flat Clustering by Embeddings*. This will allow you to partition your concordance into lines with high similarity scores, as shown in [Step 5](#).

4. Zooming in: Select by Token-Level String Attribute

Background:

Select by token-level string attribute creates a **subset** of the concordance. This means that the result will only include lines fulfilling the criteria specified by the algorithm settings.

The criteria imposed by this algorithm are based on the characteristics of the words in a specific position relative to the node. Most often, this algorithm will be used to specify a particular word immediately before or after the node term.

To specify the position, we use the **offset** parameter, where 0 corresponds to the node. Negative offsets pertain to the left context, positive offsets to the right context.

The *positional attribute* is typically set to **word**, which represents the literal word-form, or token, itself (see [4.1](#) for an example). If you have token-level annotations like POS tags (see [3.1](#) for adding POS annotations), you can also choose them as a selection criterion (see [4.2](#)).

! Note: The **checkboxes** below the offset menu offer additional settings that may be less common:

- **Case-sensitive** matching: *her* and *Her* are counted as different tokens.
- **Regex** matching: the value is interpreted as a regular expression. Using this option, you could specify that the token on offset position -1 has to end in *-ing* by selecting *+ing\b*.
- **Inverting** the selection: select all lines where the token on offset position -1 is **not** *her*.

4.1 Selecting based on the word form

Background: *Select by token-level string attribute* based on the **word** is by far the most common use-case of this algorithm: lines are selected if the word in a certain offset position corresponds to the values we entered in the *string values* window.

In the following example, we select concordance lines where the token immediately to the left of the node (= the body part noun) is *her*. The word *her* ('string attribute') is specified as a string value, and the position -1 corresponds to the 'token offset'.

As a result, the subset will contain all uses of *her cheeks*, *her cheek*, *her neck*, *her fingers*, and *her ear*.

Instructions:

1. Find *Select by token-level string attribute* in the *add algorithm* menu (Figure 12)
2. Set the **value** to match against to **her**
3. The **token attribute** is set to **word** by default – keep it that way.
4. Set the **offset** to **-1**

Find and use *Select by Token-Level String Attribute* like this:

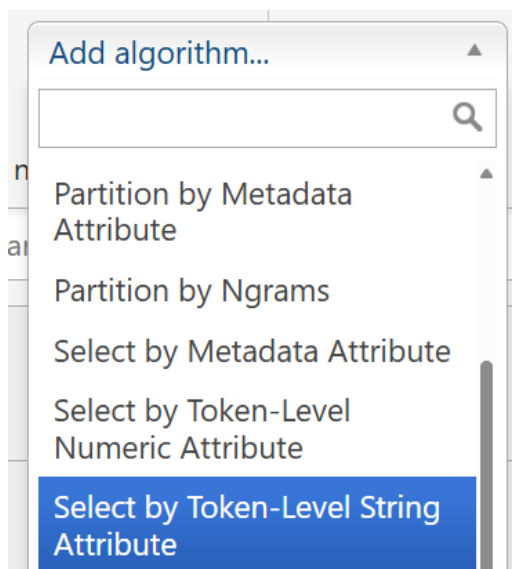


Figure 12: *Select by Token-Level String Attribute*

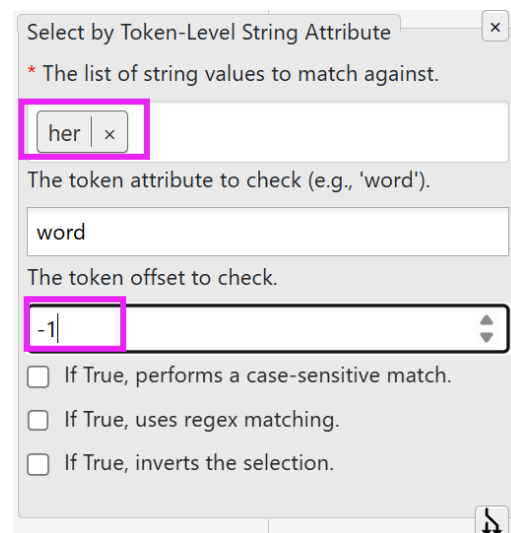


Figure 13: Parameter settings for *Select by Token-Level String Attribute* - word

This is what your result should look like:

change over the counter, in taking which he caught her fingers	and held them. There was a slight struggle and titter
the music was her only refuge from herself. Sometimes her fingers	touched the notes with a lingering fondness--a soft, plaintive
a round full face and a high colour in her cheeks.	The second sister was Fanny, a very pretty little girl
took out her handkerchief, and wiped the spot on her cheek	that had been touched by his lips. His ardour was
she began to sob, and the tears rolled down her cheeks--	a form of agitation which she had never shown before
flush with which warmth and sound sleep had suffused her cheeks	and neck dissolved from view, and the deathlike pallor in
Hale imagined that he saw traces of tears on her cheeks.	But that could not be, for she brought good news
indeed, thy fate!' whispered again the Egyptian's voice in her ear.	'And thou art destined to be the bride of Arbaces
ask me why her breath is sweet and why her cheek	is blooming, It is because the sun is out and
played caressingly with her silken hair--his breath fanned her cheek--	lone, the cruel lone, was not by--none other demanded

Figure 14: *Select by Token-Level String Attribute*: concordance of *her* on token offset -1

4.2 Selecting based on the POS tag

Background: Selecting by token-level string attributes other than the word form (token) itself is very similar to the step described above: Instead of specifying that the word in a certain position must correspond to a form like *her*, we can restrict the same position to a category such as *adjective*.

To do this, you simply need to change the positional attribute to the POS annotation (see section [3.1](#) for how to annotate your concordance with POS tags).

We will now use the spaCy tags added in [step 3.1](#) to select concordance lines based on grammatical information. In the example, we create a subset where the word immediately left to the node (offset -1) is an adjective (ADJ). As a result, we will get lines with *cold fingers*, *rosy cheeks* etc.

In the following example, we select concordance lines where the token immediately to the left of the node (= the body part noun) is *her*. The word *her* ('string attribute') is specified as a string value, and the position -1 corresponds to the 'token offset'.

As a result, the subset will contain all uses of *her cheeks*, *her cheek*, *her neck*, *her fingers*, and *her ear*.

! Note: If you are following the worked example in this document, you will currently be working on a subset of the concordance limited to *her* on position -1. Applying this additional subset to the same lines will result in an empty subset, as there cannot be lines that feature both *her* and ADJ in the same position. Therefore, you'll need to **create a new branch** (see [section 6](#)) before applying this second selection algorithm!

This new branch will be based on the full initial concordance, so that we can create a new subset.

Instructions:

1. Select a new algorithm **Select by Token-Level String Attribute** in the *add algorithm* menu (Figure 12).
2. Set the **value** to match against to **ADJ**
3. Set the **token attribute** to **pos_**
4. Set the **offset** to **-1**

Find and use Select by Token-Level String Attribute like this:

Select by Token-Level String Attribute

* The list of string values to match against.

ADJ

The token attribute to check (e.g., 'word').

pos_

The token offset to check.

-1

☐ If True, performs a case-sensitive match.

☐ If True, uses regex matching.

☐ If True, inverts the selection.

Figure 15: Parameter settings for *Select by Token-Level String Attribute* – POS tag

This is what your result should look like:

were of that sunless, flaxen hue which generally accompanies pink	cheeks	and white eyelashes. She had been a rustic beauty, perhaps
for company, and they ate together and rubbed each other's	cheeks	and brows and noses together, while they ate, with a
with a pair of amiable blue-gray eyes and round pink	cheeks	that refused to look formidable, let him frown as he
she was, the girl's Roman blood rushed over her pale	cheeks	as she felt the insult. She turned towards Hermanric, looked
She started suddenly, a bright flush flew over her colourless	cheeks;	she bent down, and looked earnestly and wistfully into the
the Prefect, staring round him with bewildered eyes and colourless	cheeks.	'Our miseries are but dawning as yet! In the next
of the stranger was of a ghastly paleness; his hollow	cheeks	were seamed with deep wrinkles; and his eyes glared with
and a deadly paleness overspread the once plump and jovial	cheeks	which so many mistresses had kissed in mercenary rapture in
tresses drooped heavily on either side of the white hollow	cheeks.	'Behold!' cried Vetrano, pointing to the corpse--'Behold my secret
days. The cold fingers that trembled round mine--the pale	cheeks	with a bright red spot burning in the midst of

Figure 16: *Select by Token-Level String Attribute*: concordance of adjectives on token offset -1

5. Investigating similarity: Flat Clustering by Embeddings

Background: Clusters are groups of concordance lines that are more similar to one another than to the remaining lines in the concordance. In the clustering methods FlexiConc supports in CLiC, the analyst determines how many groups are formed. The clustering algorithm produces precisely that number of concordance groups so that similar lines go together.

To perform clustering, you first need to annotate the concordance with one of the available embedding methods (cf. [section 3.2](#)). The similarity scores resulting from this annotation serve as the basis for the clustering algorithm to form groups.

! Notes:

1. If you are following the running example, add this algorithm to **branch 1** – that is, the lines selected for *her* on -1.
2. In the example, we use k-means clustering. FlexiConc also offers agglomerative clustering. In our experiments so far, k-means appears to work better, as the resulting clusters are more similar in size. That said, clustering for concordance lines is not yet a common approach and agglomerative clustering is worth exploring. For the agglomerative method, you can set the *distance metric* and *linkage* criterion. Under the hood, FlexiConc uses the clustering algorithms implemented in scikit-learn.

On the scikit-learn website, you can find more details about how the different clustering algorithms work: <https://scikit-learn.org/stable/modules/clustering.html>.

Instructions:

1. Select a new algorithm **Flat clustering by embeddings** in the *add algorithm* menu (Figure 17).
2. Set the **metadata column** to **embeddings_sentence_transformers** (Figure 18). If you cannot find this option, add the annotation as shown in [step 3.2](#).
3. Change the **number of clusters** to **10**. This setting determines how many groups are formed – changing it can strongly influence the results (Figure 18).

As a result of this step, all occurrences of *her* + node will be partitioned into 10 clusters of similar concordance lines.

Find and use Flat Clustering by Embeddings like this:

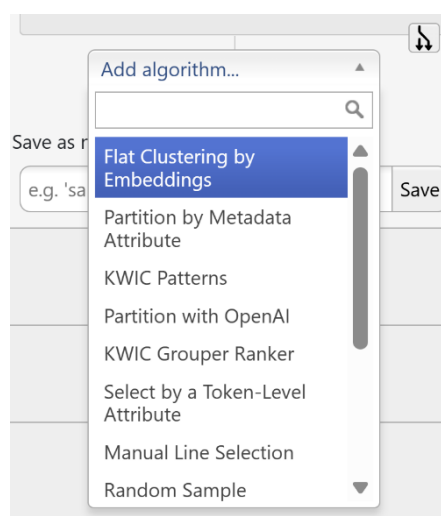


Figure 17: *Flat Clustering by Embeddings*

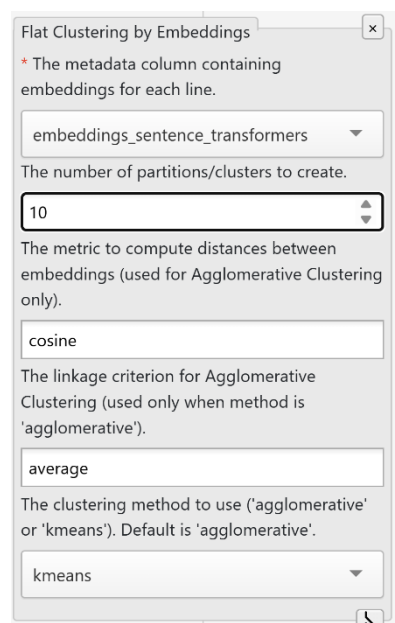


Figure 18: *Flat Clustering by Embeddings: Parameter settings*

This is what your result should look like:

On running the steps above, your concordance will be *partitioned* into clusters. By default, the first cluster (called cluster_0) is shown, with the remaining clusters collapsed (see Figure 19).

Left	Node	Right	Book	In bk.
	Partition cheeks cheek neck fingers ear Cluster_0			56 lines
quiet and still for some minutes. But, on stroking her	cheek	in a caressing way soon after, he was almost shocked	NorthS	
detect her raising a hand, and brushing something off her	cheek.	I gazed round for a means of diverting her thoughts	wh	
by her long hair and drove the knife into her	neck.	The moaning accents of the girl, bewailing her approaching fate	Antoni	
a more cheerful way, lifting her hands backward to her	neck	and moving toward the door. She wanted to waive those	Deronda	
held by the hand, was now being flung over her	neck,--	all this yeasty mingling of dimly understood facts with vague	Deronda	
foreseen that it would end with throttling fingers on her	neck.	After a pause, she said in the tone rather of	Deronda	
girl, as she spoke, touched a talisman suspended round her	neck.	"Thine own danger!" what danger threatens me? "I 'Had the earthquake	Pomp	
half-angry gesture, and twisting it backward and forward between her	fingers.	"I think some people are born to be unlucky, Mrs	LadyAud	
become upon the now familiar way! "I My lady twisted her	fingers	in her loose amber curls, and made as if she	LadyAud	
watch in her hand with the many-doubled chain round her	fingers,	and observed to Mrs. Tulliver, who had just returned from	mill	

Figure 19: Clustering view: Cluster 0 uncollapsed

Click on the partition label (marked in Figure 19) to go to an overview of all clusters (Figure 20).

In the overview of clusters (Figure 20), the number to the right shows the number of lines contained in each cluster. This shows us that clusters 0 and 1 are significantly larger than cluster 3.

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	Node	Right	Book	In bk.
Partition	cheeks cheek neck fingers ear	Cluster_0		56 lines
Partition	cheeks cheek neck fingers ear	Cluster_1		50 lines
Partition	cheeks cheek neck fingers ear	Cluster_2		37 lines
Partition	cheeks cheek neck fingers ear	Cluster_3		22 lines
Partition	cheeks cheek neck fingers ear	Cluster_4		39 lines
Partition	cheeks cheek neck fingers ear	Cluster_5		22 lines
Partition	cheeks cheek neck fingers ear	Cluster_6		44 lines
Partition	cheeks cheek neck fingers ear	Cluster_7		49 lines
Partition	cheeks cheek neck fingers ear	Cluster_8		49 lines
Partition	cheeks cheek neck fingers ear	Cluster_9		34 lines

Figure 20: Overview of cluster partitions

Clustering is a fully automated method, and the results may vary in how useful they are. On the other hand, clusters can help to identify complex patterns in the data.

Clicking on clusters 2 (Figure 21) and 3 (Figure 22) shows that both contain mostly lines for *cheeks* – but with a markedly different usage pattern:

Partition	cheeks cheek neck fingers ear	Cluster_2		37 lines
chair, with the big tears ready to roll down her	cheeks,	quite blinded to the presence of Bob, who was looking	mill	
feet were uncovered; the tears still streamed over her	cheeks.	She advanced with the greatest caution (as if fearful of	Antoni	
m her eyes. The tears streamed incessantly down her	cheeks,	but she never turned towards Goisvintha, never suspended her occupation	Antoni	
use. No words escaped her, no tears flowed over her	cheeks	when she reascended the steps--hope, fear, thought, sensation itself	Antoni	
of MY hopes too?" ¶ The tears flowed fast down her	cheeks.	She rested one trembling hand on the table to steady	wwhite	
ew tears gathered in her eyes, and dropped over her	cheeks	slowly as she paused and waited for his answer. He	wwhite	
the silence, I think; for the tears rolled down her	cheeks	like rain, at these words, and she could not answer	cran	
wiping a few tears away that had come on her	cheeks	she knew not how, she went out to open the	NorthS	
Hale imagined that he saw traces of tears on her	cheeks.	But that could not be, for she brought good news	NorthS	
tter tears of mortification rising and rolling down her	cheeks.	No one had ever before dared to treat her with	Deronda	
ithout speaking, and the tears were rolling down her	cheeks	as she looked at her friends. ¶ "Now, now, Mab!" said	Deronda	
she began to sob, and the tears rolled down her	cheeks--	a form of agitation which she had never shown before	Deronda	

Figure 21: Cluster 2 from the worked example

Partition	cheeks cheek neck fingers ear	Cluster_3	22 lines
adroit little suck to the interior of each of her	cheeks	in succession, by which curious and original manoeuvre she brought	Jude
with conflict. It did not bring the blood to her	cheeks,	but it sent it away from her lips. She controlled	Deronda
oy." ¶ Gwendolen's lips were almost as pale as her	cheeks;	her passion had no weapons--words were no better than	Deronda
uch like young animals, and so she could rub her	cheek	against his, and kiss his ear in a random sobbing	mill
rother's; baby's little fat hand, pressed against her	cheek,	seemed to show more strikingly that the cheek was faded	mill
in a close embrace, to imprint one kiss on her	cheek	or forehead? I was not unreasonable--that was all I	Prof
gure, her long white throat, and the curves of her	cheek	and chin were always set off to perfection by the	Deronda
g down by her with circling arms, she pressed her	cheek	against Gwendolen's head, and then tried to draw it upward	Deronda
ad vanity. One day, indeed, he had kissed not her	cheek	but her neck a little below her ear; and Gwendolen	Deronda
many changes besides those in the contour of her	cheek	and throat; and that Grandcourt should marry her had become	Deronda
Sybil rested on the arm of her chair, and her	cheek	upon her hand; as Egremont said these words she shaded	sybil
er agitated countenance, a tear coursed down her	cheek	big as the miserable Medora's! ¶ CHAPTER VIII. ¶ On the evening	VivianG
ib and kissed her: her forehead was cold, and her	cheek	both cold and thin, and so were her hand and	jane

Figure 22: Cluster 3 from the worked example

6. Branching out: moving in the analysis tree

Background:

The analysis tree is a key feature of FlexiConc. It documents all the algorithms we apply. The tree provides two key advantages over traditional concordance tools:

- **Research documentation:** the tree tracks all steps of your analysis, and the tree itself can be shared and restored (see [section 7](#)).
- **Supporting the analysis procedure:** the tree lets you return and build on previous steps in the analysis: see the example in this section.

The tree can be very useful during the analysis. We can quickly return to a previous step and take alternative routes by creating new branches. For instance, sections 4.1 and 4.2 contain two selection algorithms based on different criteria that were created by branching off from the same node in the tree.

In the present example, we create a branch to examine *ADJ + body-part noun* in the same way that we selected concordance lines for *her + body-part noun*. This section shows you how to create a branch from an existing node. The selection step itself is shown in [4.2](#).

! Note: The FlexiConc integration in CLiC offers two distinct ways to create branches. Both produce equivalent results, but they are suited for slightly different purposes:

- 1) The **arrow branch symbol** (Figure 23) produces a new branch that includes a copy of all steps in your old branch up to the step from which you ‘branched off’. Any algorithms performed in the active branch below the node from which you branch off (in our case, *flat clustering by embeddings* from [section 5](#)) are not copied over from the ‘old’ branch. This branching technique lets you carry over selected steps and lets you explore ‘alternative angles’.
- 2) The **plus symbol** (Figure 24) creates a fully empty top-level branch without any algorithms. This lets you “start from scratch” in a new branch while keeping the old branches intact.

Regardless of how you create a new branch, you can always switch to your previous branch by clicking 1 next to the tree symbol (Figure 24).

Instructions:

1. **Scroll up** to the node *Random sort* with the seed 42.
2. **Click** on the **branch** icon – the two crossing arrows in the bottom right corner below the seed field.
3. Follow the instructions in [4.2](#) to add a new selection based on POS tags using *Select by a token-level string attribute*.

Create and view a new branch like this:

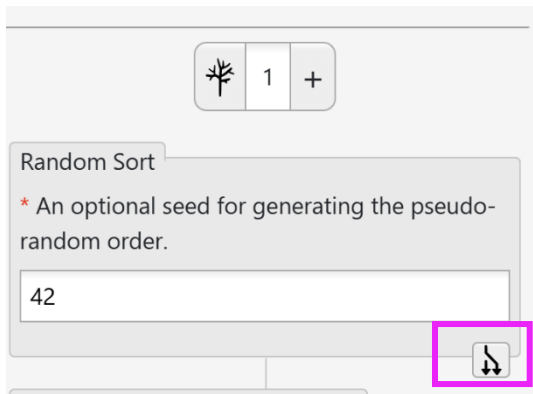


Figure 23: Branch symbol

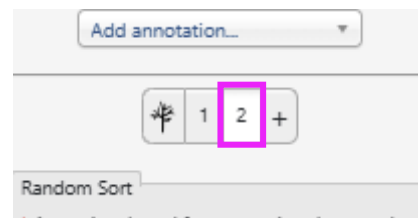


Figure 24: Your newly created branch shows up as "2"

This is what your result should look like:

This step creates a **new branch** where all steps before the one you branched off from are the same.

! Note: When using FlexiConc **within the same concordance** analysis, all your steps are automatically stored in the analysis tree.

However, if you

- change your concordance to a different query, or
- leave the website,

your steps will not be saved by default.

Therefore, **you should always save your work** when you are done ([step 16](#)).

7. The big picture: tree view

Background:

If you have been following the entire example, you now have **two branches** in your analysis.

The first branch was created when you applied the very first algorithm, Random Sort, in [section 2](#). The second branch was created in [section 6](#), when you selected by POS tag as shown in [section 4.2](#).

As the analysis becomes increasingly complex, it is useful to look at the **tree structure** as a whole. The tree view mode visualises all steps across all branches.

Instructions:

1. Click on the **tree symbol** (Figure 25) to see a visualization of the entire analysis (Figure 26).
2. In the tree overview (Figure 26), **click on 2** to switch to branch 2 (containing lines with *adjectives* to the left of the node).

Access the tree view like this:

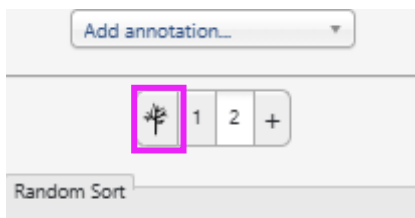


Figure 25: Tree Symbol

This is what your result should look like:

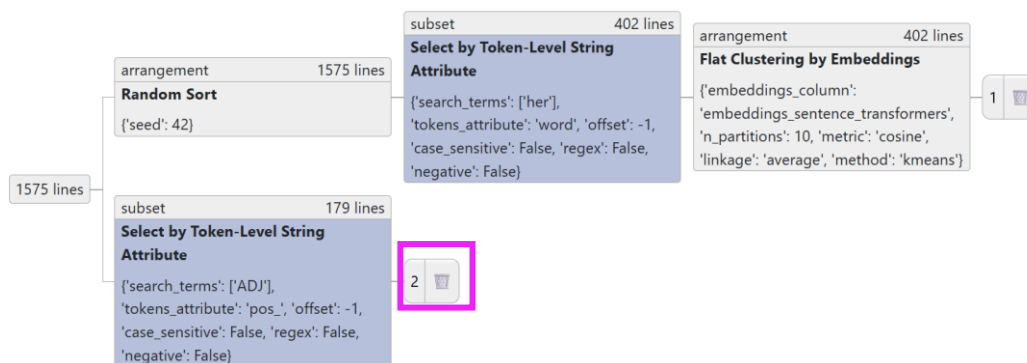


Figure 26: Analysis Tree

! Note:

The tree (Figure 26) shows all steps taken so far:

- The first box, **1575 lines**, represents the initial query.
- Branch 1 consists of three nodes:
 - The second box, **Random Sort**, represents the first algorithm.
 - Random Sort is followed by **Select by Token Level String Attribute** to select *her* on position -1.
 - The selected lines were then partitioned via **Flat Clustering by Embeddings**.
- Branch 2 contains a single node, **Select by Token Level String Attribute** with pos_ set to ADJ on -1. That is, we applied the same algorithm type in the same position twice, but on different branches.

By clicking on the numbered boxes at the end of each branch, you can switch back to that stage of the analysis.

8. Rearranging Alphabetically: Sort by Token-Level Attribute

Background:

Alphabetical sorting is one of the main methods to identify concordance patterns.

In FlexiConc, the corresponding algorithm is *Sort by Token-Level Attribute*. Like *Select by Token-Level Attribute* ([section 4](#)), sorting can be applied to the token itself (default) or to any annotation on the token layer (like [POS tags, section 3.1](#)) after annotating the concordance.

In branch 2 of our example, we selected lines based on the POS tag on the position -1. As a result, the token left to the node can be any word that was tagged as an adjective.

The words that precede the node in this subset share a grammatical category, but the word form itself will differ between lines.

To see whether these adjectives repeat, we can use alphabetical sorting on the word level.

Instructions:

1. Select the algorithm *Sort by Token-Level Attribute* (Figure 27).
2. Set the *offset* to -1 – the position one token to the left of the node (Figure 28).

! Note:

By changing the *context* setting in the second drop-down menu, you could also sort on the entire left or right context (= everything before/ after the node).

For our present case, however, sorting on a specific token is more useful to see which adjectives are repeated (e.g. *ashy cheeks*, see *Figure 27*).

Find and use Sort by Token-Level Attribute like this:

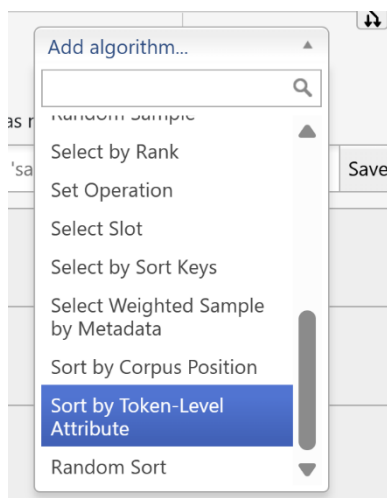


Figure 27: Sort by Token-Level Attribute

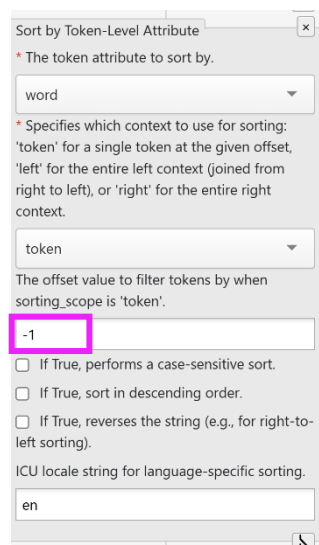


Figure 28: Parameter settings for Sort by Token-Level Attribute

This is what your result should look like:

Showing 1 to 50 of 179 entries,

ID	Left	Node	Right
1451	herself crouching in her hitherto pure waters, while her anxious	ear	listens for the bounding step of the regardless huntsman. ¶ "The
102	from his mother rose dark and slow in his ashy	cheeks.	¶ "Is that woman," he asked, "the woman whom you once
189	the narcotic. Again some return of colour to the ashy	cheeks,	and the regular breathing of healthy sleep. This time I
1274	of the rapid Tiber now caught his suspicious and attentive	ear.	They still moved onward a few yards; and then paused
1446	bending over her with starting eyes, once more the attentive	ear	listening for the soundless breath. No sound! not even a
611	up and struck furiously with the knife at her bare	neck.	As the weapon descended, Hermanric caught her wrist. She struggled
99	look, aided by his light hair and his plump beardless	cheeks,	his easy manner and his ever-ready smile, his eyes which
425	were a tender inflexibility, the tears trickling down her beautiful	cheek,	and pressing his hand in both of hers, she subdued
1184	have!" Mr. Osborne said to his clerk, snapping his big	fingers,	and jingling all the guineas and shillings in his great
112	observed that a warm flush now rosed her hitherto blue	cheeks,	and felt that her hanging hand was no longer cold
519	with her golden ringlets flying loose behind, and her bright	cheek,	as soft and pure in its bloom as a wild
301	left no obvious trace; the eyes are liquid, the brown	cheek	is firm and round, the full lips are red. With
973	responsively at every casual movement of the usher's supple brown	fingers,	and every passing distortion of the usher's haggard yellow face
1236	the bridge. And that queer white cur with the brown	ear	seems to be leaping and barking in ineffectual remonstrance with
875	tenderly coaxing the crumbled leaf of his cihar with cautious	fingers.	¶ "My friend at the corner of Chancery Lane has not

Figure 29: Concordance with ADJ on offset -1, sorted alphabetically on -1.

9. Groups based on sequences: Partition by N-grams

Background:

N-grams are sequences of a fixed length. By setting n to a numeric value, you can change that length to one that suits your research question. A common choice are bigrams ($n = 2$) and trigrams ($n = 3$), as these often point to useful patterns across different texts.

In FlexiConc, n-grams are offered as a **partitioning** algorithm, which creates groups with frequency counts, similar to [section 5](#).

While our example below shows traditional n-grams, you can also use this function to create **skip-grams** (e.g. you could choose positions -1 and 1 to see repeated patterns of preceding and following tokens irrespective of the node term).

As is the case with *Select* and *Select by Token-Level Attribute* ([sections 4](#) and [8](#)), you can also use annotations like POS ([section 3.1](#)) to create n-grams. To do this, change the *word* attribute to the token-level annotation layer that you want to use (pos_ for POS tags).

In this example, we use word-level 2-grams (also known as *bigrams*) to get a more comprehensive overview of the adjectives used on L1 and the body part noun used in the node.

! Note:

In the CLiC interface outside of FlexiConc mode, n-grams are called *clusters*. We decided to use different terminology to avoid overlap with our clustering algorithm ([step 5](#)).

Instructions:

1. Select the algorithm **Partition by N-grams** (Figure 30).
2. Add **-1** and **0** to the list of positions to create bigrams for the word on -1 and the node (Figure 31).

Find and use Partition by N-grams like this:

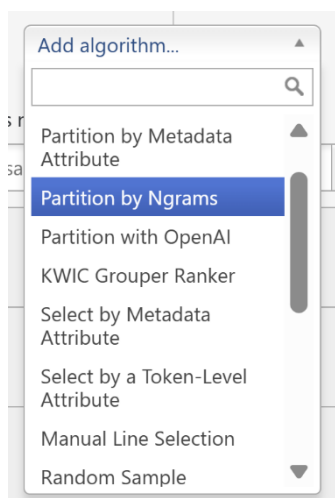


Figure 28: *Partition by N-grams*

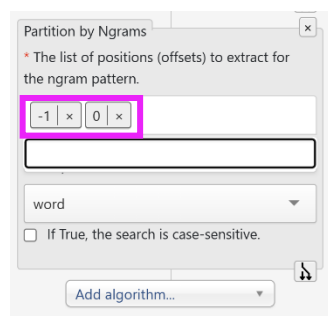


Figure 29: *Partition by N-grams*: Parameter settings

This is what your result should look like:

Partition	cheeks cheek neck fingers ear	('pale', 'cheeks')	9 lines
she was, the girl's Roman blood rushed over her pale cheeks	as she felt the insult. She turned towards Hermanric, looked	Antoni	
lays. The cold fingers that trembled round mine--the pale cheeks	with a bright red spot burning in the midst of	wwhite	
inted palace to their appreciative taste--Sue's usually pale cheeks	reflecting the pink of the tinted roses at which she	Jude	
way. I in love--and with him too! Her pale cheeks	suddenly became one flame of fire; and she covered her	NorthS	
now was sunken pallor. But the children kissed the pale cheeks	and never found them deficient. That love was now the	Deronda	
'ou saw Miss Harleth?' 'I " _She_ told you that?' 'I The pale cheeks	looked even paler, perhaps from the fierce brightness in the	Deronda	
the fresh breeze had braced her up, and her pale cheeks	were really more rosy. Her mother rejoiced when she saw	dracula	
through the window and gave more colour to the pale cheeks.	When he saw her, Arthur was simply choking with emotion	dracula	
tears, and caused them to fall quietly, down her pale cheeks.	'f It was the longest day she had ever known in	mary	
Partition	cheeks cheek neck fingers ear	('little', 'fingers')	6 lines
Partition	cheeks cheek neck fingers ear	('pink', 'cheeks')	5 lines
Partition	cheeks cheek neck fingers ear	('pale', 'cheek')	4 lines
Partition	cheeks cheek neck fingers ear	('small', 'fingers')	4 lines

Figure 30: Concordance with *ADJ* on position -1, partitioned by n-grams

10. Choosing what matters: Manual Line Selection

Background:

With *Manual Line Selection*, you can create a subset based on lines that you choose. This is especially useful for partitions, as Manual Line Selection lets you pick out one partition to analyse it in more detail.

In the example below, the manual line selection is based on the bigrams created using *Partition by Ngrams* (see [section 9](#)). We will use manual selection to group lines together that share some token positions, but are not entirely the same. We will focus on partitions that indicate cheek complexion. For each partition, the token on -1 is an adjective like *pale* and the node is *cheek* or *cheeks*.

Manual line selection has two modes:

1. you can select **line IDs** to select individual lines, or
2. **group identifiers** (the label or number of the partitions to choose).
In our example, we will use this second option to select the entire partition.

Instructions:

1. Select the algorithm **Manual line selection** (Figure 31).
2. **Click on the group identifier window** (Figure 32).
3. Choose the partitions 'pale cheeks,' 'pink cheeks,' and 'pale cheek' (Figure 33).
4. Confirm by clicking **OK** in the bottom left (Figure 33).

Find and use Manual Line Selection like this:

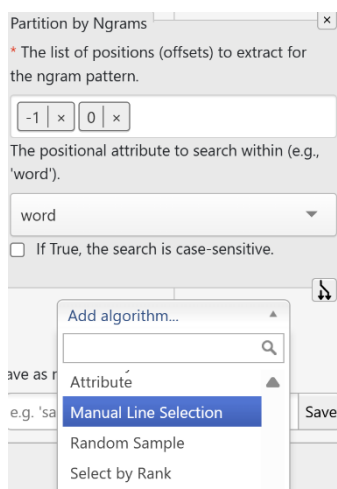


Figure 31: *Manual Line Selection*

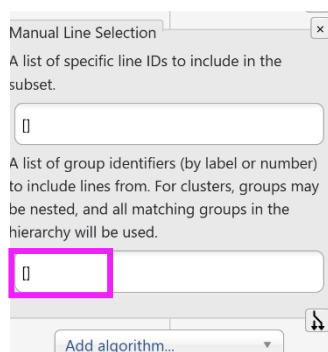


Figure 32: *Manual Line Selection: Parameter settings*

Select the partitions like this:

ID	Left	Node	Right	Book	In bk.
▼ ☒	Partition	cheeks cheek neck fingers ear	('pale', 'cheeks')		9 lines
35	she was, the girl's Roman blood rushed over her pale cheeks		as she felt the insult. She turned towards Hermanric, looked	Antoni	☐
60	ays. The cold fingers that trembled round mine--the pale cheeks		with a bright red spot burning in the midst of	wwhite	☐
114	enchanted palace to their appreciative taste--Sue's usually pale cheeks		reflecting the pink of the tinted roses at which she	Jude	☐
143	way. I in love--and with him too! Her pale cheeks		suddenly became one flame of fire; and she covered her	NorthS	☐
160	now was sunken pallor. But the children kissed the pale cheeks		and never found them deficient. That love was now the	Deronda	☐
161	you saw Miss Harleth?" "I "She told you that?" "I The pale cheeks		looked even paler, perhaps from the fierce brightness in the	Deronda	☐
185	the fresh breeze had braced her up, and her pale cheeks		were really more rosy. Her mother rejoiced when she saw	dracula	☐
190	through the window and gave more colour to the pale cheeks.		When he saw her, Arthur was simply choking with emotion	dracula	☐
270	tears, and caused them to fall quietly, down her pale cheeks.		"It was the longest day she had ever known in	mary	☐
▶ ☐	Partition	cheeks cheek neck fingers ear	('little', 'fingers')		6 lines
▶ ☒	Partition	cheeks cheek neck fingers ear	('pink', 'cheeks')		5 lines
▼ ☒	Partition	cheeks cheek neck fingers ear	('pale', 'cheek')		4 lines

Figure 33: Manually selecting partitions

This is what your results should look like:

ID	Left	Node	Right	Book	In bk.
▼	Partition	cheeks cheek neck fingers ear	('pale', 'cheeks')		9 lines
35	she was, the girl's Roman blood rushed over her pale	cheeks	as she felt the insult. She turned towards Hermanric, looked	Antoni	
60	days. The cold fingers that trembled round mine--the pale	cheeks	with a bright red spot burning in the midst of	wwhite	
114	an enchanted palace to their appreciative taste--Sue's usually pale	cheeks	reflecting the pink of the tinted roses at which she	Jude	
143	way. I in love--and with him too! Her pale	cheeks	suddenly became one flame of fire; and she covered her	NorthS	
160	now was sunken pallor. But the children kissed the pale	cheeks	and never found them deficient. That love was now the	Deronda	
161	You saw Miss Harleth? "I ". She told you that?" "I The pale	cheeks	looked even paler, perhaps from the fierce brightness in the	Deronda	
186	the fresh breeze had braced her up, and her pale	cheeks	were really more rosy. Her mother rejoiced when she saw	dracula	
190	through the window and gave more colour to the pale	cheeks.	When he saw her, Arthur was simply choking with emotion	dracula	
270	tears, and caused them to fall quietly, down her pale	cheeks.	"It was the longest day she had ever known in	mary	
►	Partition	cheeks cheek neck fingers ear	('pink', 'cheeks')		5 lines
►	Partition	cheeks cheek neck fingers ear	('pale', 'cheek')		4 lines

Figure 34: Manually selected partitions

11. A bit of everything: Random Sample

Background: Random samples are useful when

- the concordance is too large to analyse fully, or
- if you want to generate hypotheses before investigating them in the overall corpus

In [section 2](#), we applied Random Sort, which retained all lines in the concordance and showed them in a semi-random order.

In contrast, Random Sample performs random sorting in the background and then creates a subset with the number of lines specified by the user.

Instructions:

1. Create a new branch in the tree using the + button in the branch overview (Figure 35).
2. Select the **Random Sample** algorithm. (Figure 36).
3. Set the **number of lines to sample** to **500** (Figure 37).
4. Keep the **random seed**, which ensures a reproducible sample (Figure 37).

! Notes:

1. You should always set a **seed** to make your random sampling reproducible.
2. Creating a **branch** via the + sign results in a *top-level branch*. None of the algorithms used so far are carried over – branch 3 is an entirely blank slate, while branches 1 & 2 are left intact.

This is how to find and use *Random Sample*:

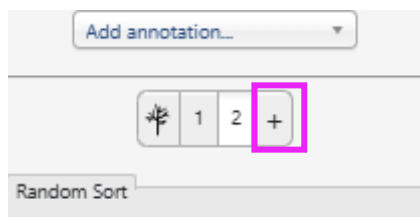


Figure 35: Creating a new top-level branch

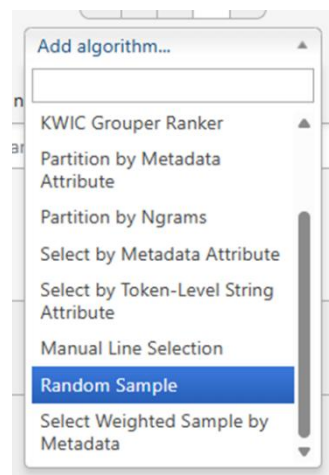


Figure 36: *Random Sample*

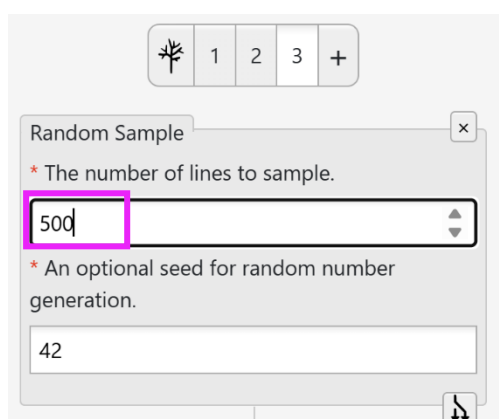


Figure 37: *Random Sample*: Parameter settings

This is what your result should look like:

Showing 1 to 50 of 500 entries,

ID	Left	Node	Right	Book	In bk.
1	at half-past nine o'clock, singing a little Scotch melody, her	cheeks	tinged with as delicate a pink as the pale hue	LadyAud	
3	of that sunless, flaxen hue which generally accompanies pink	cheeks	and white eyelashes. She had been a rustic beauty, perhaps	LadyAud	
4	clinging about him, kissed the dirty tears from his faded	cheeks.	"Don't be sorry for me, gran'pa," he said; "I am	LadyAud	
6	better nor half the folks as are grown up." "Maggie's	cheeks	began to flush with triumphant excitement. She thought Mr. mill		
7	lavishing so many warm kisses on it that the waxen	cheeks	had a wasted, unhealthy appearance. "Did you ever hear t: mill		
13	and still Tom laughed and clapped his hands, and Maggie's	cheeks	began to pale, and her lips to tremble a little	mill	
14	turn towards the same point as her own, and Maggie's	cheeks	and ears began to burn, while uncle Glegg, a kind-looking	mill	
16	t Columbus?" "Maggie's eyes had begun to sparkle and her	cheeks	to flush,--she was really beginning to instruct the gypsies	mill	
19	ne ambitious. The exercise brought the warm blood into her	cheeks,	and made her inclined to take her lesson merrily. "I	mill	
23	gone before, a glowing gladness at heart. Her eyes and	cheeks	were still brightened with her childlike enthusiasm in the da: mill		

Figure 38: Concordance of a *Random Sample* with 500 lines

This handout builds on work funded by the "Reading Concordances in the 21st Century" research project supported by the Arts and Humanities Research Council (AHRC) (grant references: AH/X002047/1 & AH/X002047/2) and the Deutsche Forschungsgemeinschaft (DFG) (grant reference: 508235423).

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12. Information about the text: Select by Metadata Attribute

Background:

Metadata is information *about* the text rather than what is *in* the text (the concordance lines).

Which metadata attributes are available and what they are called may vary between corpora.

In the 19C corpus, available metadata includes:

- the book title (coded as **text_id**)
- chapter number, and
- further structural information.

Like Select by Token-Level String Attribute ([section 4](#)), Select by Metadata Attribute creates a subset of the original concordance. As we are dealing with a corpus of novels, Select by Metadata Attribute lets us focus on concordance lines in specific books.

Instructions:

1. Find *Select by Metadata Attribute* in the *add algorithm* menu (Figure 39).
2. Set the **metadata attribute** to **text_id** (Figure 40).
3. Set the **value to compare against** to **Deronda** (Figure 41).

Find and use Select by Metadata Attribute like this:

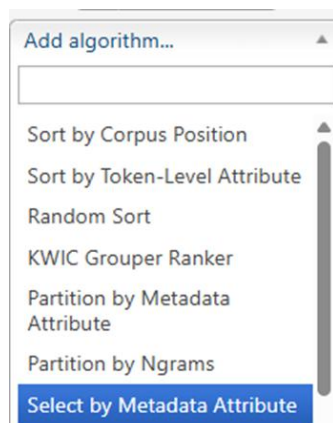


Figure 39: *Select by Metadata Attribute*

Select by Metadata Attribute

* The metadata attribute to filter on.

text_id

* The value to compare against, or a list of acceptable values. When a list is provided, only equality is used.

Deronda x

The comparison operator for numeric comparisons. Only allowed for single numeric values. Default is '='.

==

☐ If True, use regex matching for string comparisons (only with equality). Default is False.

☐ If True, perform case-sensitive matching for strings. Default is False.

☐ If True, invert the selection. Default is False.

Figure 40: *Select by Metadata Attribute: Parameter settings*

This is what your result should look like:

Showing 1 to 45 of 45 entries,

ID	Left	Node	Right	Book	In bk.
148	with conflict. It did not bring the blood to her	cheeks,	but it sent it away from her lips. She controlled	Deronda	
150	longed to somebody else. Her eyes had become brighter, her	cheeks	slightly flushed, and her tongue ready for any mischievous re	Deronda	
154	followed hers. The handsome little fellow was puffing out his	cheeks	in trying to blow a tiny trumpet which remained dumb	Deronda	
161	You saw Miss Harleth?" "She_ told you that?" "The pale	cheeks	looked even paler, perhaps from the fierce brightness in the	Deronda	
163	of his pockets; or drawing down the skin of his	cheeks	to make his eyes look awful, and rolling his head	Deronda	
164	stool without speaking, and the tears were rolling down her	cheeks	as she looked at her friends. "Now, now, Mab!" said	Deronda	
167	her son's shoulders at once, and kissed him on both	cheeks,	then seated herself among her cushions with an air of	Deronda	
392	ch by giving her sisters lessons." Here Mrs. Davilow's delicate	cheek	showed a rapid blush. "If it were not for that	Deronda	
393	figure, her long white throat, and the curves of her	cheek	and chin were always set off to perfection by the	Deronda	
395	after the long sorrow and watching which had paled her	cheek	and made blue semicircles under her eyes. It was Mab	Deronda	
403	knew you would be." Then she kissed him on each	cheek,	and he returned the kisses. But it was something like	Deronda	
407	from her foretaste of joy; some couple, bending cheek by	cheek,	over a bit of work done by the one and	Deronda	
408	She was seated gazing fixedly on the sea, resting her	cheek	on her hand, looking less shattered than when he had	Deronda	
668	in an infallible system which placed his foot on the	neck	of chance--immediately prepared a new pile. So did a	Deronda	

Figure 41: *Select by Metadata Attribute: Concordance lines from the novel Deronda*

This handout builds on work funded by the "Reading Concordances in the 21st Century" research project supported by the Arts and Humanities Research Council (AHRC) (grant references: AH/X002047/1 & AH/X002047/2) and the Deutsche Forschungsgemeinschaft (DFG) (grant reference: 508235423).

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13. Grouping by books or chapters: partition by metadata

Background:

In this step, we use **metadata** – information *about* the text rather than what is *in* the text (the concordance lines) – to group concordance lines.

Partition by Metadata creates partitions of the type that were also used in [step 5](#) (Flat clustering) and [step 9](#) (Partition by Ngrams). By partitioning based on metadata, we can see how patterns are distributed across all novels in the text, or across all chapters in a specific book. In this example, we apply chapter partitions to our *Deronda* subset from [section 12](#) to see whether body part nouns are more common in specific chapters of the novel.

Instructions:

1. Find *Partition by Metadata* in the *add algorithm* menu (Figure 42)
2. Set the metadata attribute to **chapter** (Figure 43).
3. Keep the checkbox ticked to see partitions sorted from largest to smallest (Figure 43).

Find and use Partition by Metadata Attribute like this:

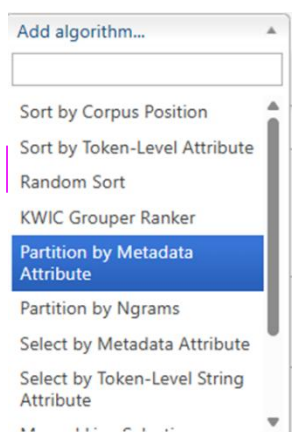


Figure 42: *Partition by Metadata Attribute*

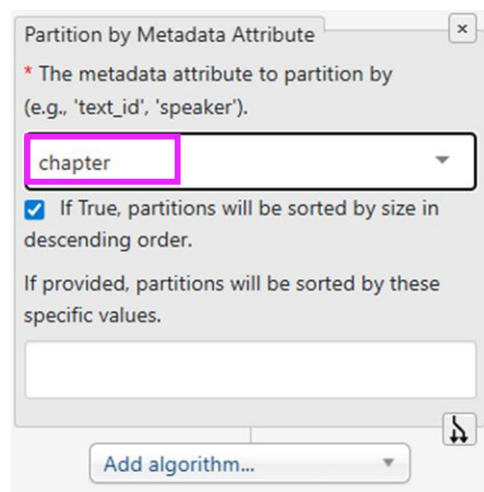


Figure 43: *Partition by Metadata Attribute: Parameter settings*

This is what your result should look like:

Partition	cheeks cheek neck fingers ear	48	12 lines
boy.* ¶ Gwendolen's lips were almost as pale as her	cheeks;	her passion had no weapons--words were no better than	Deronda
and from the yoke she had brought on her own	neck.	No! she foresaw him always living, and her own life	Deronda
that she should die with his throttling fingers on her	neck	avenging that thought. Fantasies moved within her like ghosts, making	Deronda
black dress her head on its white pillar of a	neck	showed to advantage. Some consciousness of this made her turn	Deronda
her crown of hair so as completely to conceal her	neck,	and leave only her face looking out from the black	Deronda
e, her husband was arriving with a cigar between his	fingers.	He threw it away and handed her out, accompanying her	Deronda
into the terror that she should die with his throttling	fingers	on her neck avenging that thought. Fantasies moved within her	Deronda
f with the jewels that glittered on her tightly-clasped	fingers	pressed against her heart. ¶ The feeling Deronda endured in these	Deronda
hand, and she let him press her poor little chill	fingers;	but she said no good-bye. ¶ When he had left the	Deronda
of the right shape; nor one the lobe of whose	ear	was at all too large and red; nor one who	Deronda
ile sneer in his low voice which in poor Gwendolen's	ear	was diabolical. ¶ For the first time since their marriage she	Deronda
ad no intention to the contrary, but to Gwendolen's	ear	his words had as much insolence in them as his	Deronda
Partition	cheeks cheek neck fingers ear	30	9 lines
Partition	cheeks cheek neck fingers ear	23	7 lines
Partition	cheeks cheek neck fingers ear	29	7 lines
Partition	cheeks cheek neck fingers ear	39	6 lines

Figure 44: *Partition by Metadata Attribute*: Concordance lines from the novel *Deronda* across chapters.

! Note:

As you can see in Figure 46, *Deronda* has many chapters, making it difficult to spot any tendencies right away.

💡 If you see any partitions or lines that interest you, use **Manual Line Selection** ([section 10](#)) to create different subsets from a specific range of chapters (e.g. the first and second half of the novel).

14. Maintaining balance: Select Weighted Sample by Metadata

Background:

The *Select Weighted Sample by Metadata* creates a subset of the concordance that contains the same proportion of lines per book or chapter as in the overall concordance.

To create this sample, the user specifies:

- the **number of lines** to draw, similar to Random Sample ([section 11](#))
- a **metadata attribute** to weight by, such as *text_id* (book title).

Compared to Random Sample, which treats all lines as equally likely, the weighted sample will create a sample that is more similar to the overall concordance in terms of the specified attribute.

! Note: As with any semi-random operation, you should use a **seed** to ensure your selection is reproducible.

Instructions:

1. Create a new empty **branch** using the plus sign (+) next to the tree symbol (Figure 45).
2. Find *Select Weighted Sample by Metadata* in the *add algorithm* menu (Figure 46).
3. Create a sample based on the attribute **text_id** (the book title, Figure 47).
4. Set the **total number of lines** to **500** (Figure 47).
5. Keep the **random seed** intact (Figure 47).

Find and use Select Weighted Sample by Metadata like this:

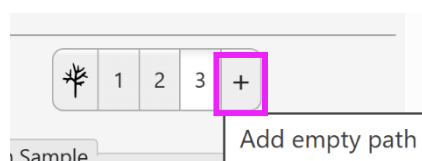


Figure 45: Create an empty branch

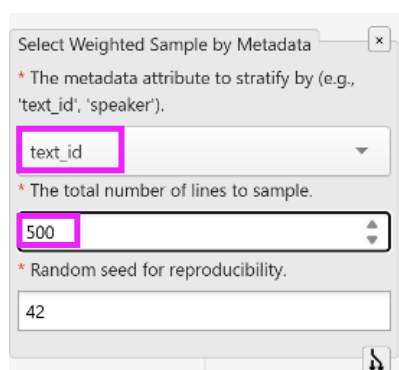


Figure 47: *Select Weighted Sample by Metadata*: Parameter Settings

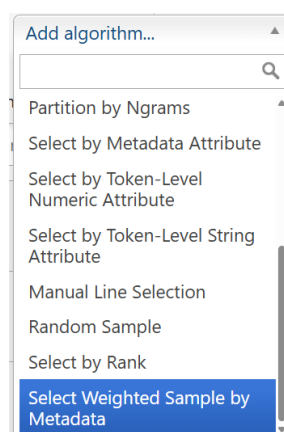


Figure 46: *Select Weighted Sample by Metadata*

This is what your result should look like:

Left	Node	Right
were of that sunless, flaxen hue which generally accompanies pink	cheeks	and white eyelashes. She had been a rustic beauty, perhaps
better nor half the folks as are grown up." ¶ Maggie's	cheeks	began to flush with triumphant excitement. She thought Mr. Riley
lavishing so many warm kisses on it that the waxen	cheeks	had a wasted, unhealthy appearance. ¶ "Did you ever hear the
turn towards the same point as her own, and Maggie's	cheeks	and ears began to burn, while uncle Glegg, a kind-looking
Maggie, my dear; don't make yourself so ugly!" and Tom's	cheeks	were looking particularly brilliant as a relief to his best
about Columbus?" ¶ Maggie's eyes had begun to sparkle and her	cheeks	to flush,--she was really beginning to instruct the gypsies
with a pair of amiable blue-gray eyes and round pink	cheeks	that refused to look formidable, let him frown as he
chair, with the big tears ready to roll down her	cheeks,	quite blinded to the presence of Bob, who was looking
was the instinctive vent of strong excitement. Her eyes and	cheeks	had an almost feverish brilliancy; her head was thrown backward
must have come again by the river. Maggie felt her	cheeks	glowing and her heart beating; it was natural she should
he had lashed himself into fury. His eyes glared, his	cheeks	flushed, his voice rose. Could he then have seen the

Figure 48: Concordance for *Select Weighted Sample by Metadata*

15. Ranking by pattern counts: KWICGrouper Ranker

Background:

KWICGrouper Ranker is a variant of KWICGrouper (O'Donnell 2008). KWICGrouper Ranker reorders the concordance lines based on how many matches of specified search terms they contain.

Compared to the KWICGrouper implementation in the traditional CLiC interface, KWICGrouper Ranker features some extra functionality, such as switching between type (every form is only counted once) and token counts (every match is sorted, regardless of repetitions).

In the example, we will rank the concordance lines based on the number of *-ing* forms.

! Note: If you have annotated your concordance with POS tags ([section 3.1](#)), you can also KWICGroup on the PoS level, for instance, to rank by the number of adjectives (ADJ).

In that case, you should disable the checkbox for type-counting to make sure that more than one ADJ tag per line can be counted.

Instructions:

1. If you are following the overall example, add this algorithm to the Weighted Random Sample created in **branch 4** (Figure 45).
2. Find *KWICGrouper Ranker* in the *add algorithm* menu (Figure 49).
3. Set the **matching strategy** to *cqp* (Figure 50)
4. Enter **.+ing** as a search term to find words that end in *ing* (Figure 50)

Find and use KWICGrouperRanker like this:

Select Weighted Sample by Metadata

* The metadata attribute to stratify by (e.g., 'text_id', 'speaker').

text_id

* The total number of lines to sample.

500

An optional seed for generating the pseudo-random order.

42

Add algorithm...

KWIC Grouper Ranker

Partition by Metadata Attribute

Save

Figure 49: KWIC Grouper Ranker

KWIC Grouper Ranker

* A list of terms to search for within the tokens.

.+ing

word

Matching strategy for search_terms

cqp

☐ If True, the search is case-sensitive.

☐ If True, include node-level tokens in the search.

The lower bound of the window (offset range).

The upper bound of the window (offset range).

☒ If True, count unique types within each line; otherwise, count all matches.

Save

Figure 50: KWIC Grouper Ranker: Parameter Settings

! Note: The search term **.+ing** that we ranked on is a **regular expression**. Regex allow you to specify patterns -ing forms regardless of which words they are. By using **cqp** mode over **regex** mode, token boundaries are automatically assumed - **.+ing** in traditional regex mode would also retrieve *thingamy* unless you specify an explicit word boundary **.+ing \$**

.+ means that the word can start with any character and be of arbitrary length; but there must be at least one character before *ing*.

\$ means that the word has to end after *-ing*, so we do not get matches like *thingamy*. We will, however, still get some unintended results (e.g. *something*).

This is what your result should look like:

not here?' she demanded, pulling off her gloves, and displaying fingers	wonderfully whitened with doing nothing and staying indoors. ¶ 'Heathcliff, you
work, and went on making false stitches and pricking her fingers	with much perseverance, not looking up or taking notice of
who was speaking with animation--now stretching out his long fingers	horizontally, now pointing downward with his fore-finger, now folding his
instant on the shadowing eye-lashes before rolling slowly down her cheeks,	and dropping , unheeded, on her dress. ¶ They were to stop
rushing into the room, threw his arms round the Prince's neck. ¶ "My darling! " said the father, forgetting at this moment of	
much for the sables," she said, holding out her little fingers,	all glittering and twinkling with the diamonds she wore upon
He looked vexed and puzzled, and sat drumming with his fingers	on the dining-table without uttering a word. ¶ "I'm afraid something
her back towards them, was counting her change with trembling fingers,	not daring to turn round until she heard the answer
spilling their milk over the table and themselves, plunging their fingers	into their own or each other's mugs, or quarrelling over
jewels from her arms, and the rings from her shaking fingers,	and held them all in a heap, quivering and looking
Mrs. Tulliver, beckoning Maggie to her, and whispering in her ear,	as soon as this point of Lucy's staying was settled
must have come again by the river. Maggie felt her cheeks	glowing and her heart beating ; it was natural she should
like to hear it." ¶ Midwinter suddenly looked up again, his cheeks	turning ashy pale, and his glittering black eyes fixed full
spoke these last words the tears were running down his cheeks,	and he was weeping openly in presence of his companion

Figure 51: Concordance for *KWICGrouperRanker*

! Note: KWICGrouperRanker creates a new column **Ranking**, which shows the number of matches for each concordance line: the first line contains 5 -ing words, followed by two lines with four matches (Figure 52).

	Book	In bk.	Ranking: KWIC Grouper Ranker
wh			5
mill			4
Deronda			4
NorthS			3
VivianG			3
LadyAud			3
arma			3
NorthS			3
AgnesG			3
vanity			3

Figure 52: Rank score for *KWICGrouperRanker*

16. Wrapping up: saving and loading your work

Background:

Besides being useful for analysis, the tree that FlexiConc creates ([section 6](#) and [section 7](#)) serves reproducibility and long-term transparency.

You can save your tree to a JSON file and load the same file into FlexiConc at a later stage. This allows you to:

- pause your analysis and resume it later.
- share your work with a colleague.
- append your analysis to a journal article or term paper.
- trace other analysts' steps to understand their procedure.

Instructions:

1. Go to the **analysis tree** view by clicking on the tree symbol (Figure 53).
2. **Click on save to file.** While in tree view mode, you can find this button right below the tree symbol (Figure 54).
3. The tree is automatically downloaded. Find it in your downloads folder and name it **rc21_flexiclic_demo.json**
4. Click on load from file and select the tree file for upload. FlexiConc should start reloading your analysis.

This is how to find the tree and the save/load functions

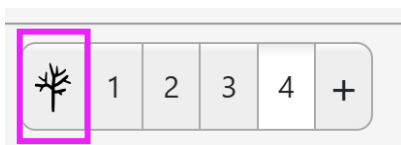


Figure 53: Accessing the analysis tree.

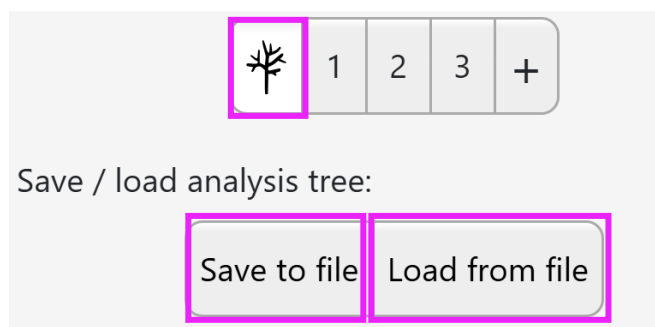


Figure 54: Save and load buttons.

This is what your tree should look like:

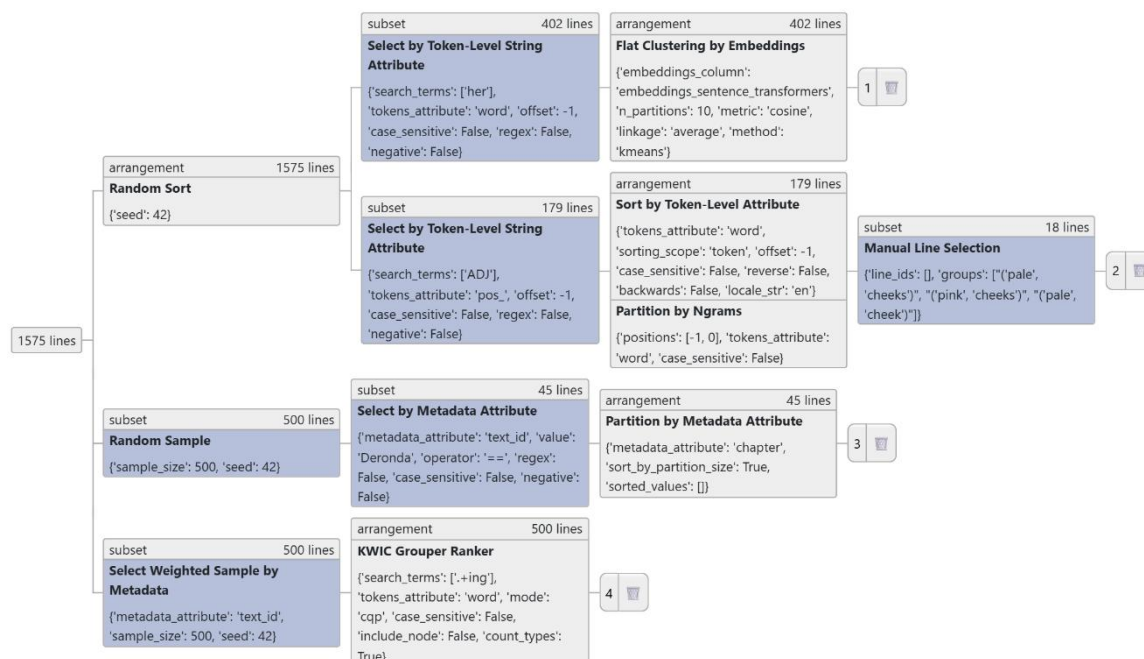


Figure 55: Analysis tree

If you open the tree on your hard drive using a text editor, you will see something similar to this, depending on which editor you use:

```
{
  "doc": "/flexiconc",
  "args": {
    "corpora": {
      "corpus": "19C"
    },
    "conc-subset": {
      "tokens": {
        "tokens": "any"
      },
      "tokens_attribute": {
        "tokens_attribute": "any"
      },
      "tokens_offset": {
        "tokens_offset": "any"
      },
      "tokens_case_sensitive": {
        "tokens_case_sensitive": "any"
      },
      "tokens_regex": {
        "tokens_regex": "any"
      },
      "tokens_negative": {
        "tokens_negative": "any"
      },
      "tokens_case_sensitive": {
        "tokens_case_sensitive": "any"
      },
      "tokens_regex": {
        "tokens_regex": "any"
      },
      "tokens_negative": {
        "tokens_negative": "any"
      }
    },
    "tokens_attribute": {
      "tokens_attribute": "any"
    },
    "tokens_offset": {
      "tokens_offset": "any"
    },
    "tokens_case_sensitive": {
      "tokens_case_sensitive": "any"
    },
    "tokens_regex": {
      "tokens_regex": "any"
    },
    "tokens_negative": {
      "tokens_negative": "any"
    }
  },
  "algorithms": [
    {
      "name": "Random Sort",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    },
    {
      "name": "Select by Token-Level String Attribute",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    },
    {
      "name": "Flat Clustering by Embeddings",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    },
    {
      "name": "Sort by Token-Level Attribute",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    },
    {
      "name": "Partition by Metadata Attribute",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    },
    {
      "name": "KWIC Grouper Ranker",
      "tokens_attribute": "word",
      "tokens_offset": -1,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false,
      "tokens_case_sensitive": false,
      "tokens_regex": false,
      "tokens_negative": false
    }
  ]
}
```

Figure 56: Analysis tree in JSON format

If you're unfamiliar with JSON files, don't worry – this is how FlexiConc stores the information on algorithms and settings that is needed to restore your analysis.

! By default, the file is downloaded as clic-analysis-tree. json. **You should always rename your file and move it to the folder where you keep your project!**

1. Name your tree **rc21_flexiclic_demo.json**
2. Save it in a location where you will be able to find it again!

References

- Mahlberg, M., Stockwell, P., Wiegand, V. and Lentin, J. (2020) CLiC 2.1. Corpus Linguistics in Context.
- O'Donnell, M. B. (2008). KWICgrouper: Designing a tool for corpus-driven concordance analysis. *International Journal of English Studies*, 8(1), 107-122.
- Reimers, Nils, and Iryna Gurevych. "Sentence-bert: Sentence embeddings using siamese bert-networks." *arXiv preprint arXiv:1908.10084* (2019).

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19C: Sybil Tawse, illustration for *Cranford* by Elizabeth Gaskell (1914), captioned “*She always told them to hold out their tiny palms, into which she shook either peppermint or ginger lozenges.*” Internet Archive: <https://archive.org/details/cranford191400gaskuoft>

ArTS: E.W. Kemble, *Striking for the Back Country*, in Mark Twain, *Adventures of Huckleberry Finn* (1885), p. 275. Wikimedia Commons: https://commons.wikimedia.org/wiki/File:Adventures_of_Huckleberry_Finn_1885-p275.png

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